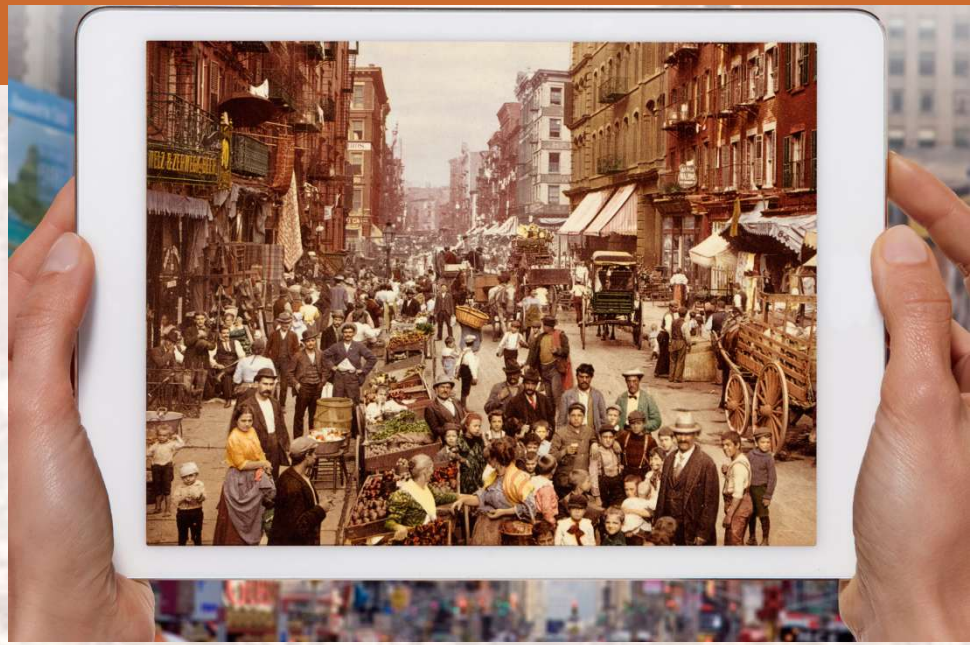
經濟學



CHAPTER 8

Application: The Costs of Taxation

National Central University

Look for the answers to these questions:

- How does a tax affect consumer surplus, producer surplus, and total surplus?
- What is the deadweight loss of a tax?
- What factors determine the size of this deadweight loss?
- How does tax revenue depend on the size of the tax?



Review from Chapter 6

- A tax
  - Drives a wedge between the price buyers pay and the price sellers receive
  - Raises the price buyers pay and lowers the price sellers receive
  - Reduces the quantity bought and sold
- These effects are the same
  - Whether the tax is imposed on buyers or sellers

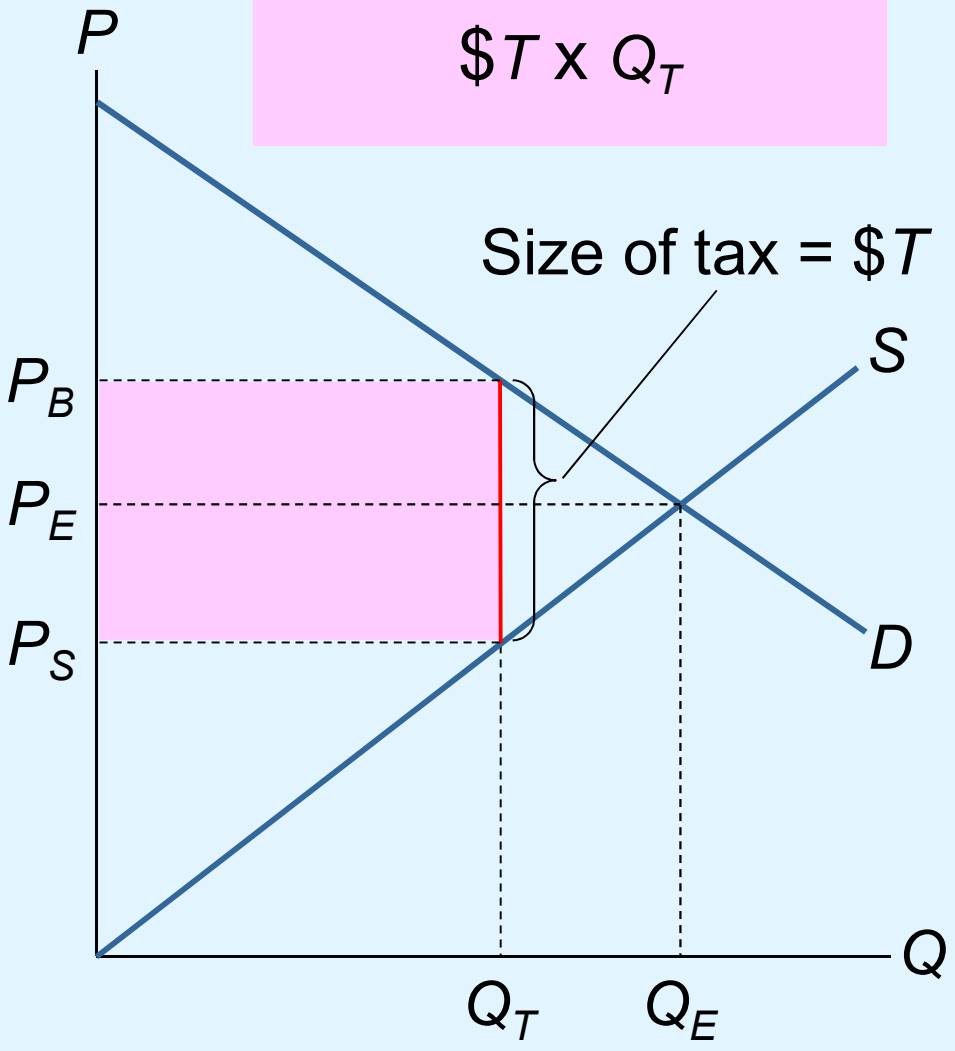
The Effects of a Tax

Equilibrium with no tax:

- Price = P_E
- Quantity = Q_E

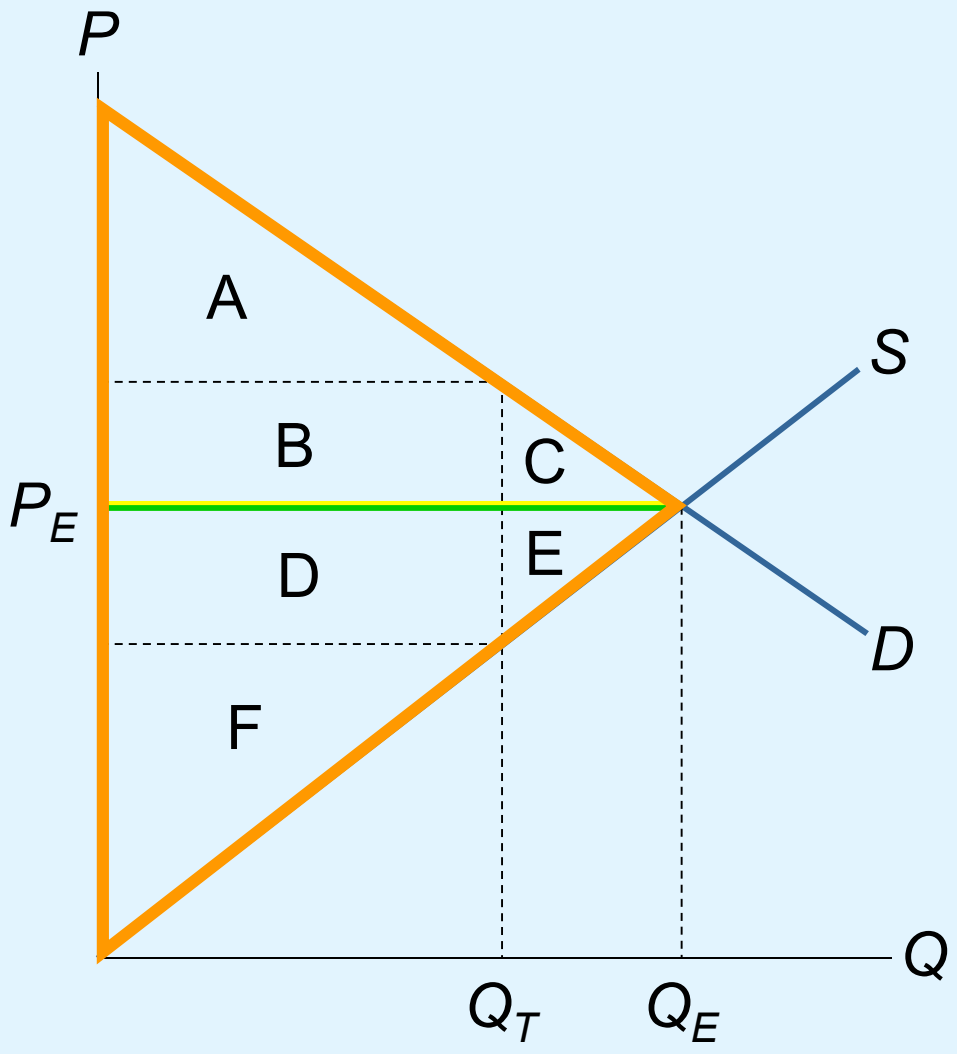
Equilibrium with tax = $T per unit:

- Buyers pay P_B
- Sellers receive P_S
- Quantity = Q_T



The Effects of a Tax

Without a tax,
CS = A + B + C
PS = D + E + F
Tax revenue = 0
Total surplus
= CS + PS
= A + B + C
+ D + E + F



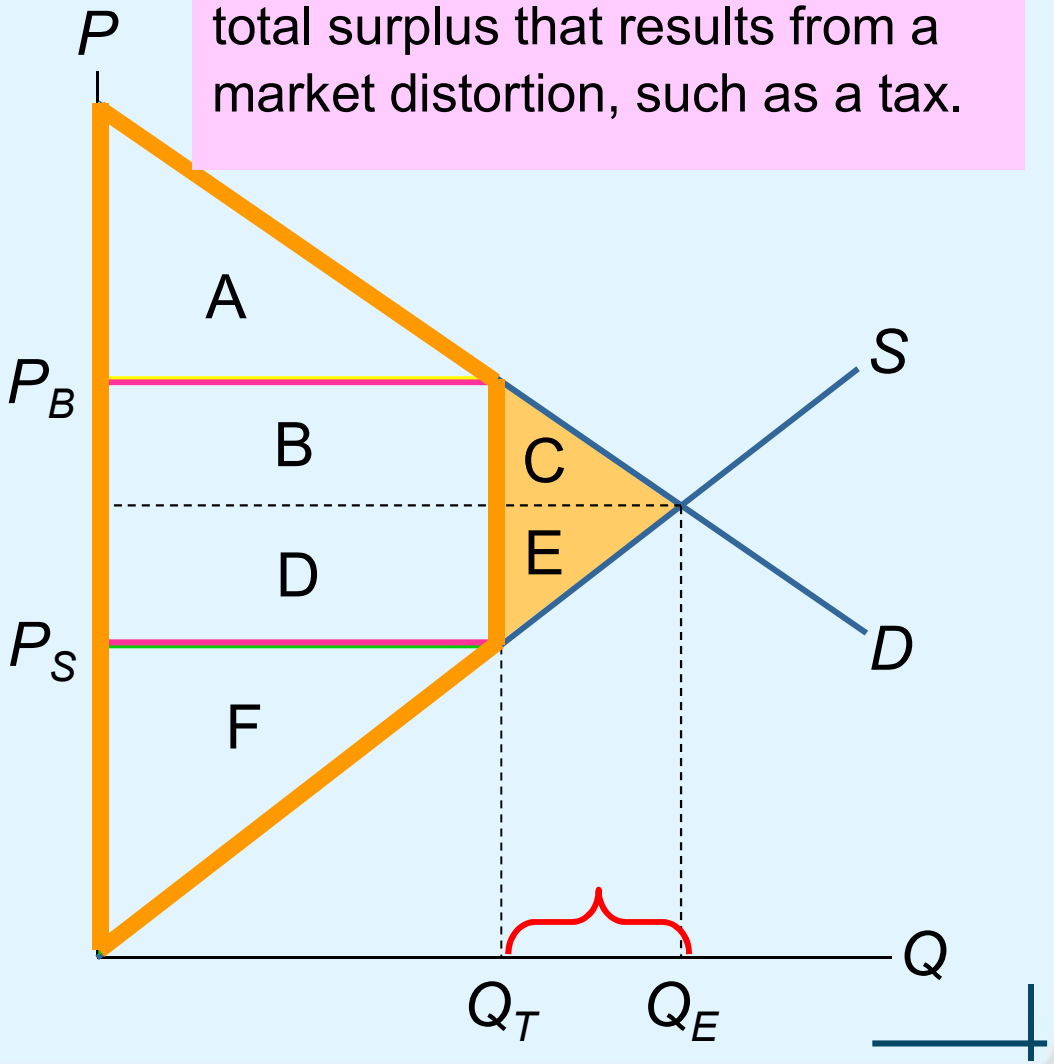
The Effects of a Tax

DWL: 無謂的損失

With the tax,
CS = A
PS = F
Tax revenue = B + D
Total surplus
= CS + PS + Tax
= A + B + D + F
The tax reduces total surplus by C + E

$Q_E - Q_T$ = units not sold because of the tax

C + E is called the **deadweight loss** (DWL) of the tax, the fall in total surplus that results from a market distortion, such as a tax.

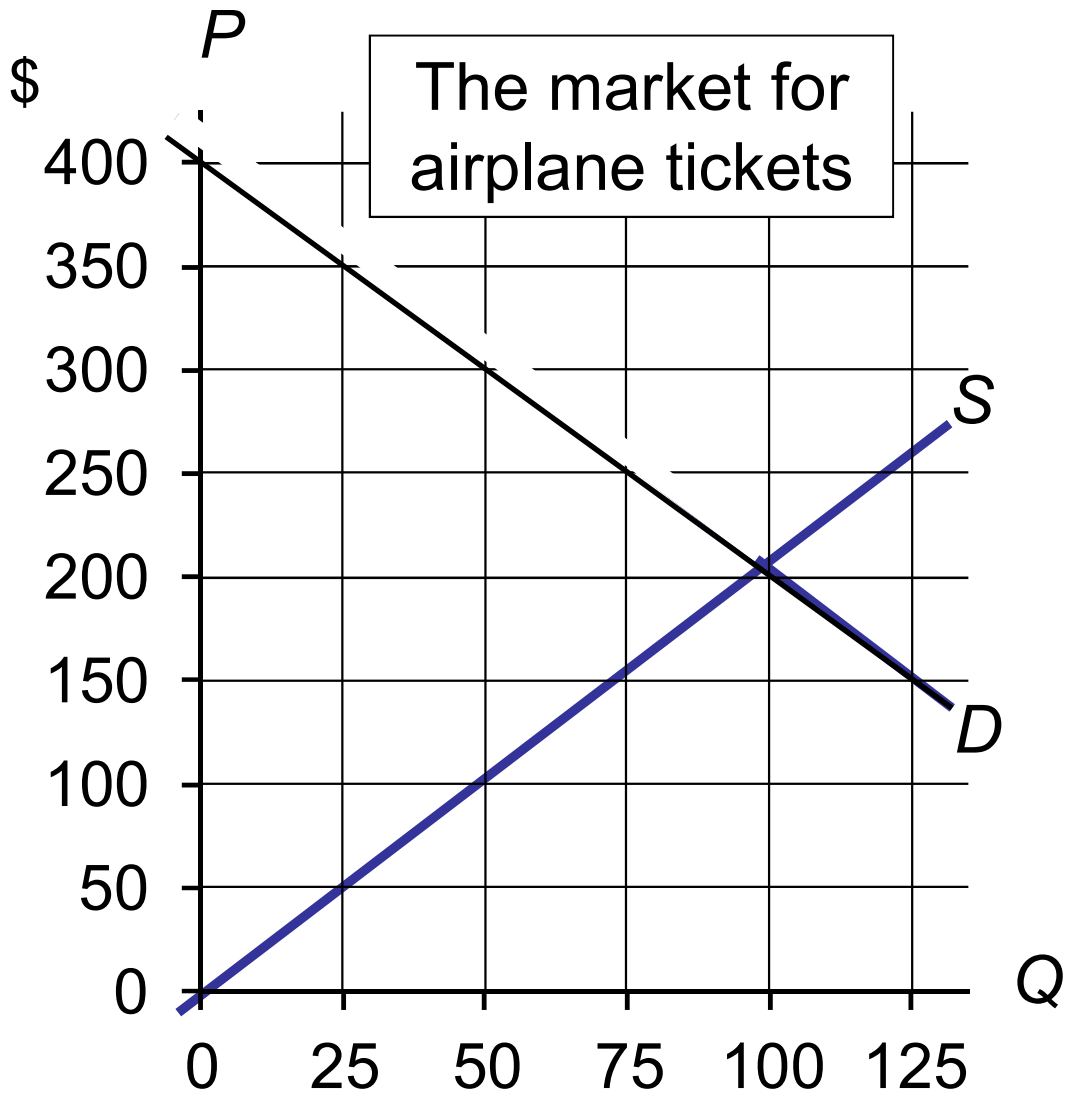


Active Learning 1

Analysis of a tax

A. Compute CS, PS, and total surplus without a tax.

B. If $100 tax per ticket, compute CS, PS, tax revenue, total surplus, and DWL.



Active Learning 1

A. Answers

A. Without tax:

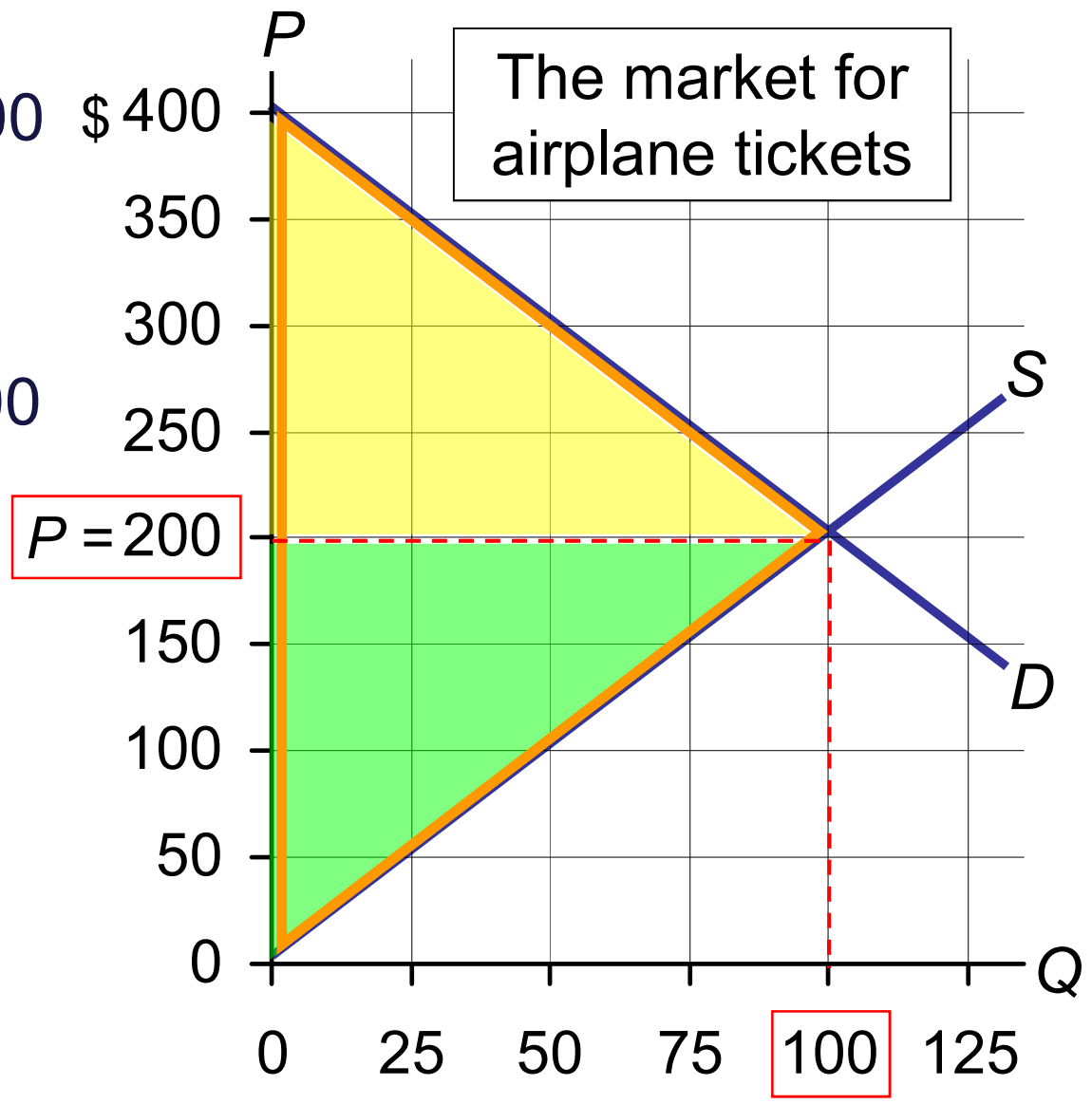
CS = ½ x \$200 x 100

= \$10,000

PS = ½ x \$200 x 100

= \$10,000

TS = \$20,000



Active Learning 1

B. Answers

B. With tax:

CS = ½ x $150 x 75

= $5,625

PS = $5,625

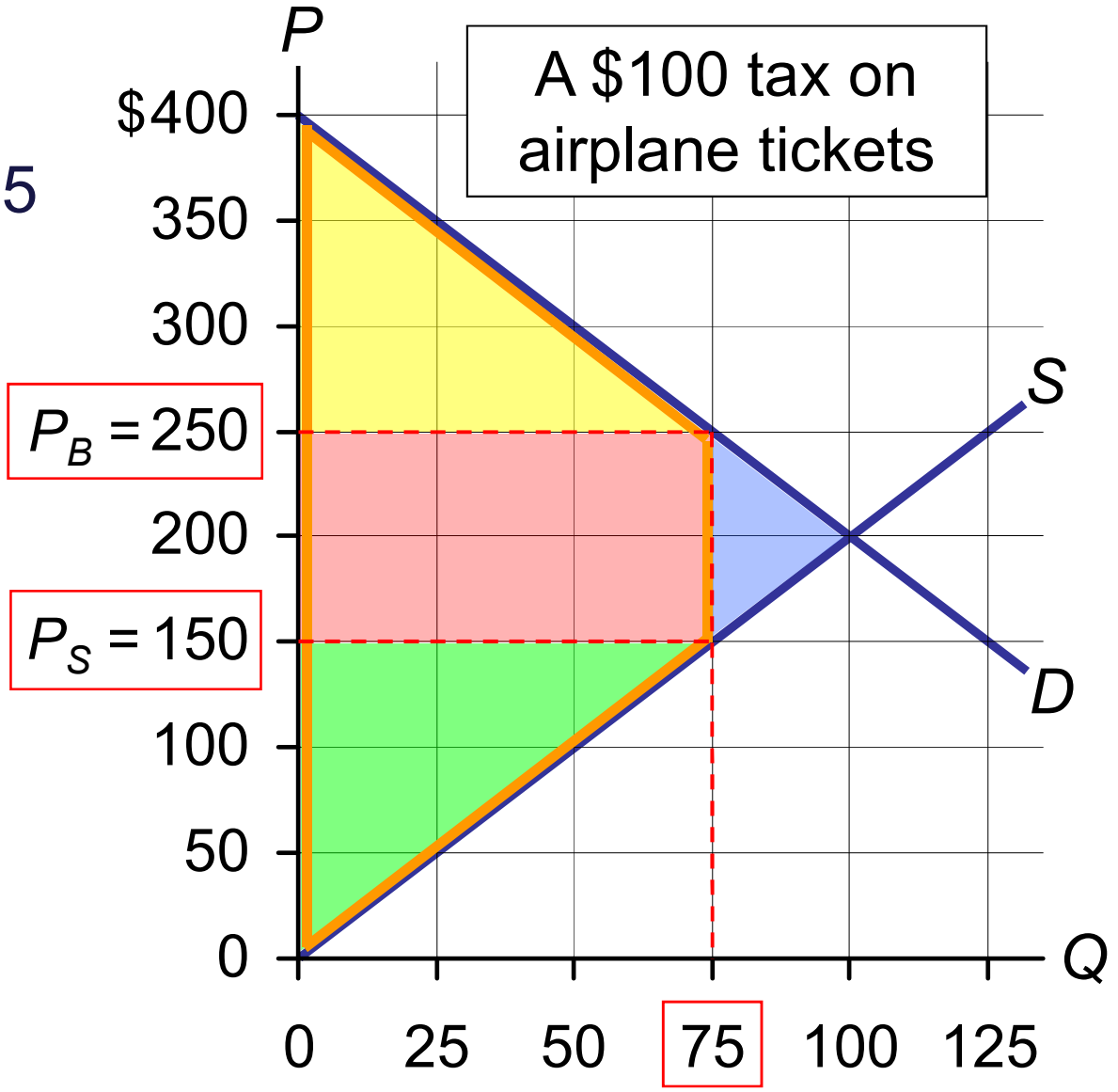
Tax revenue

= $100 x 75

= $7,500

TS = $18,750

DWL = $1,250





Determinants of Deadweight Loss

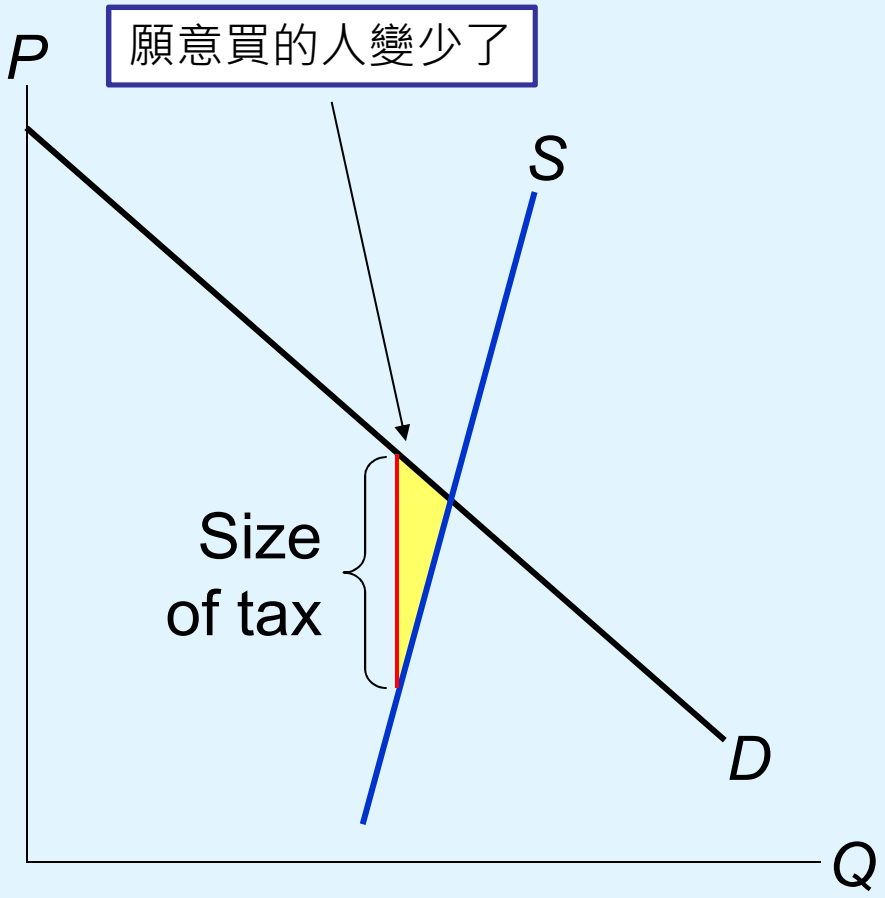
- Price elasticities of supply and demand
  - More elastic supply curve
    - Larger deadweight loss
  - More elastic demand curve
    - Larger deadweight loss
- The greater the elasticities of supply and demand
  - The greater the deadweight loss of a tax

DWL and the Elasticity of Supply

When supply is inelastic,

it's harder for firms to leave the market when the tax reduces P_S.

So, the tax only reduces Q a little, and DWL is small.

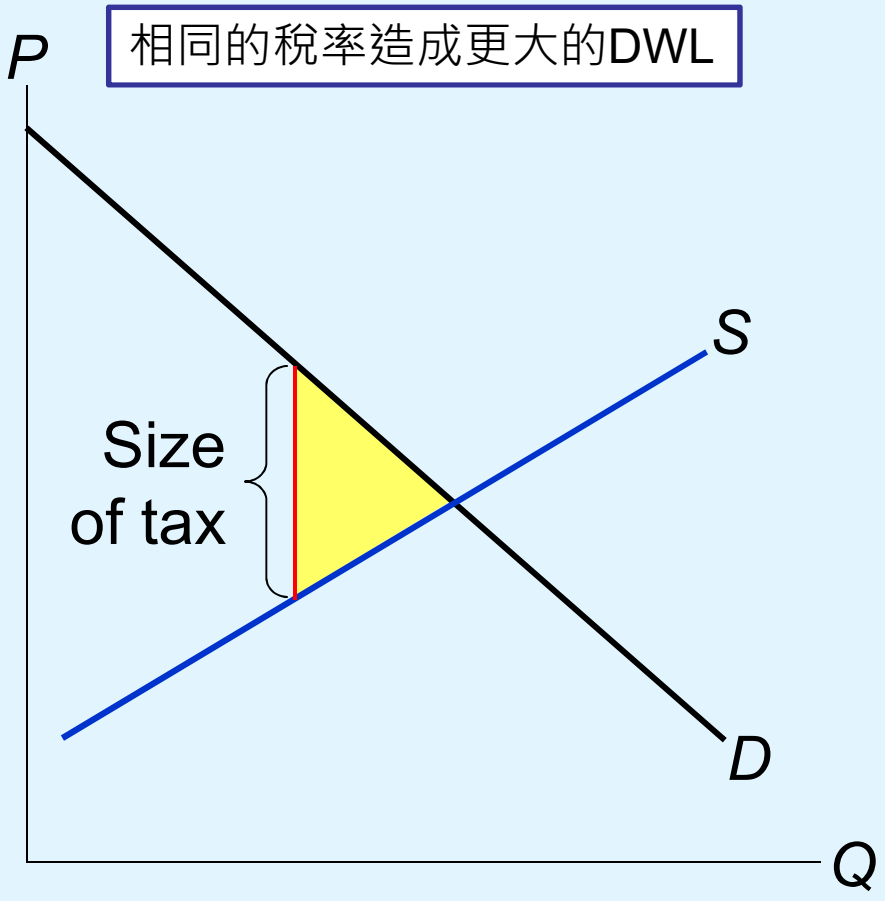


DWL and the Elasticity of Supply

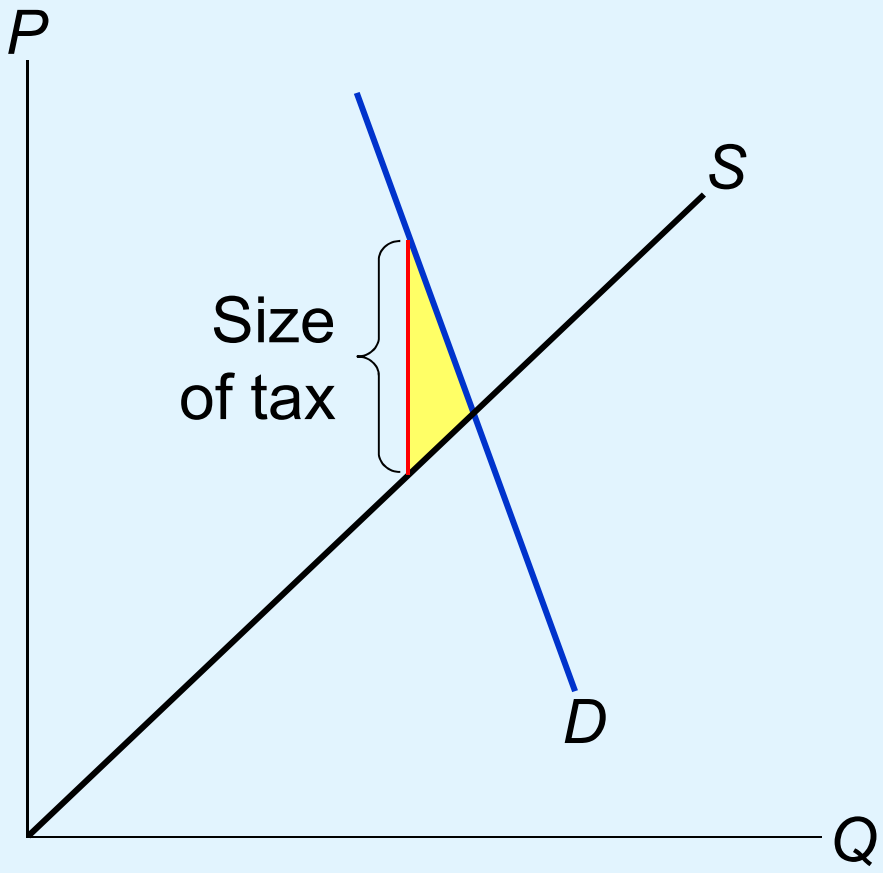
The more elastic is supply,
the easier for firms to leave the market
when the tax reduces P_S,

the greater Q falls below the surplus-maximizing quantity,

the greater the DWL.



DWL and the Elasticity of Demand

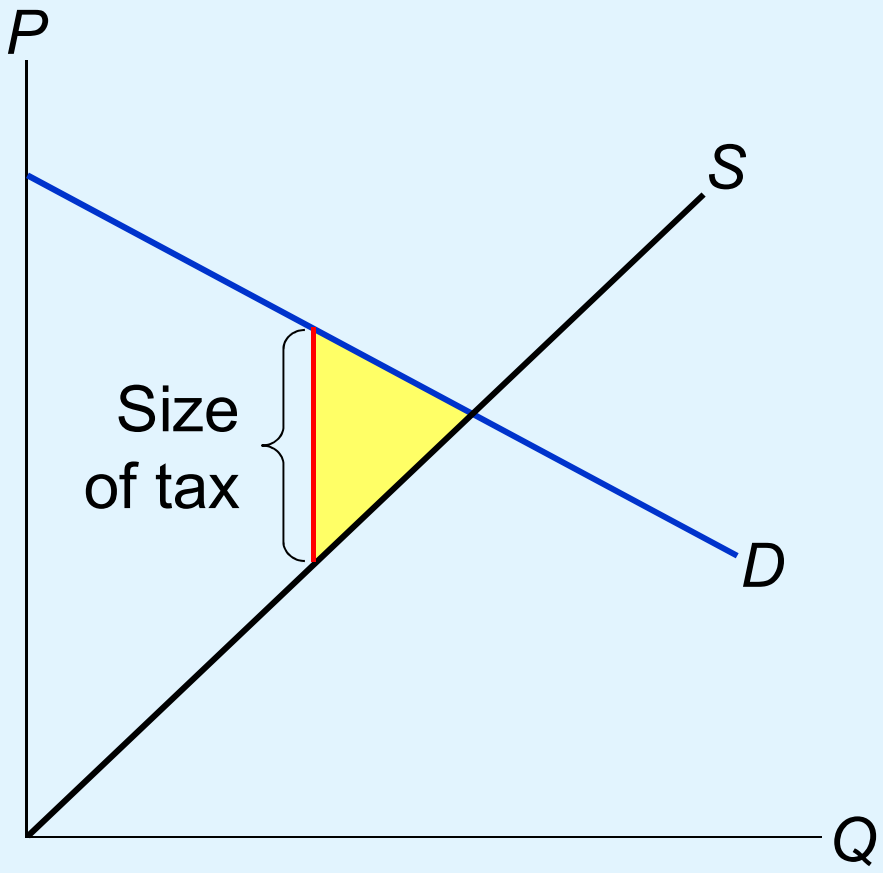


When demand is inelastic,

it's harder for consumers to leave the market when the tax raises P_B.

So, the tax only reduces Q a little, and DWL is small.

DWL and the Elasticity of Demand



The more elastic is demand,
the easier for buyers to leave the market when the tax increases P_B,

the more Q falls below the surplus-maximizing quantity,

and the greater the DWL.

Active Learning 2 — Elasticity and the DWL

Would the DWL of a tax be larger if the tax were on:

A. Breakfast cereal or sunscreen?

B. Hotel rooms in the short run or hotel rooms in the long run?

C. Groceries or meals at fancy restaurants?

Active Learning 2 Answers

A. Breakfast cereal or sunscreen?

From Chapter 5:
Breakfast cereal has more close substitutes than sunscreen, so demand for breakfast cereal is more price-elastic than demand for sunscreen.

- So, a tax on breakfast cereal would cause a larger DWL than a tax on sunscreen.

Active Learning 2 Answers

B. Hotel rooms in the short run or hotel rooms in the long run?

From Chapter 5:
The price elasticities of demand and supply for hotel rooms are larger in the long run than in the short run.

- So, a tax on hotel rooms would cause a larger DWL in the long run than in the short run.

Active Learning 2 Answers

C. Groceries or meals at fancy restaurants?

From Chapter 5:
Groceries are more of a necessity and therefore less price-elastic than meals at fancy restaurants.

- So, a tax on restaurant meals would cause a larger DWL than a tax on groceries.

Active Learning 3 — Discussion question

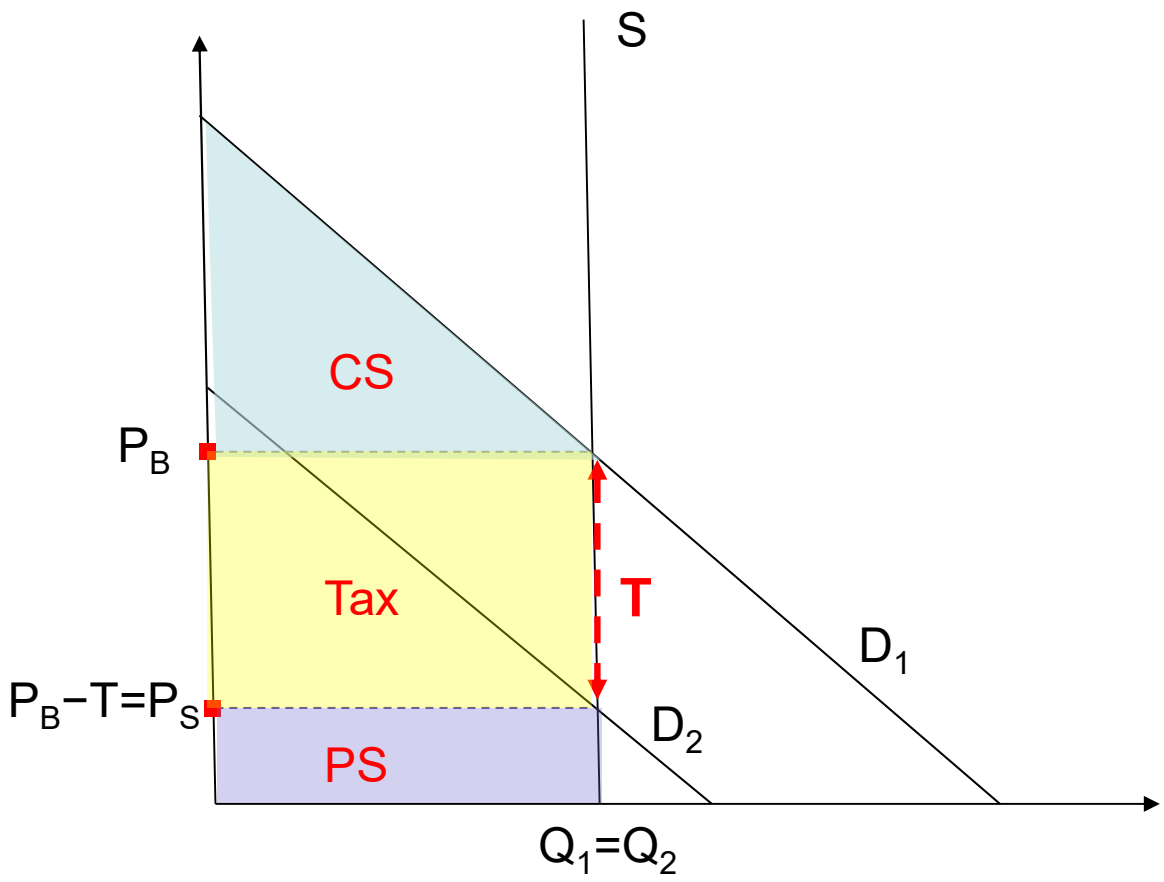
The government must raise tax revenue to pay for schools, police, etc. To do this, it can either tax groceries or meals at fancy restaurants.

- Which should it tax?

Appendix Question

- Evaluate the following two statements. Do you agree?
  - A tax that has no deadweight loss cannot raise any revenue for the government
  - A tax that raises no revenue for the government cannot have any deadweight loss

Appendix Effect of a Tax on Buyers



Appendix Question

- Consider the market for rubber bands
- If this market has very elastic supply and very inelastic demand, how should the burden of a tax on rubber bands be shared between consumers and producers?
- If this market has very inelastic supply and very elastic demand, how would the burden of a tax on rubber bands be shared between consumers and producers?



Appendix Question

- Suppose that the government imposes a tax on heating oil
- Would the deadweight loss from this tax likely be greater in the first year after it is imposed or in the fifth year?
- Would the revenue collect from this tax likely be greater in the first year after it is imposed or in the fifth year?

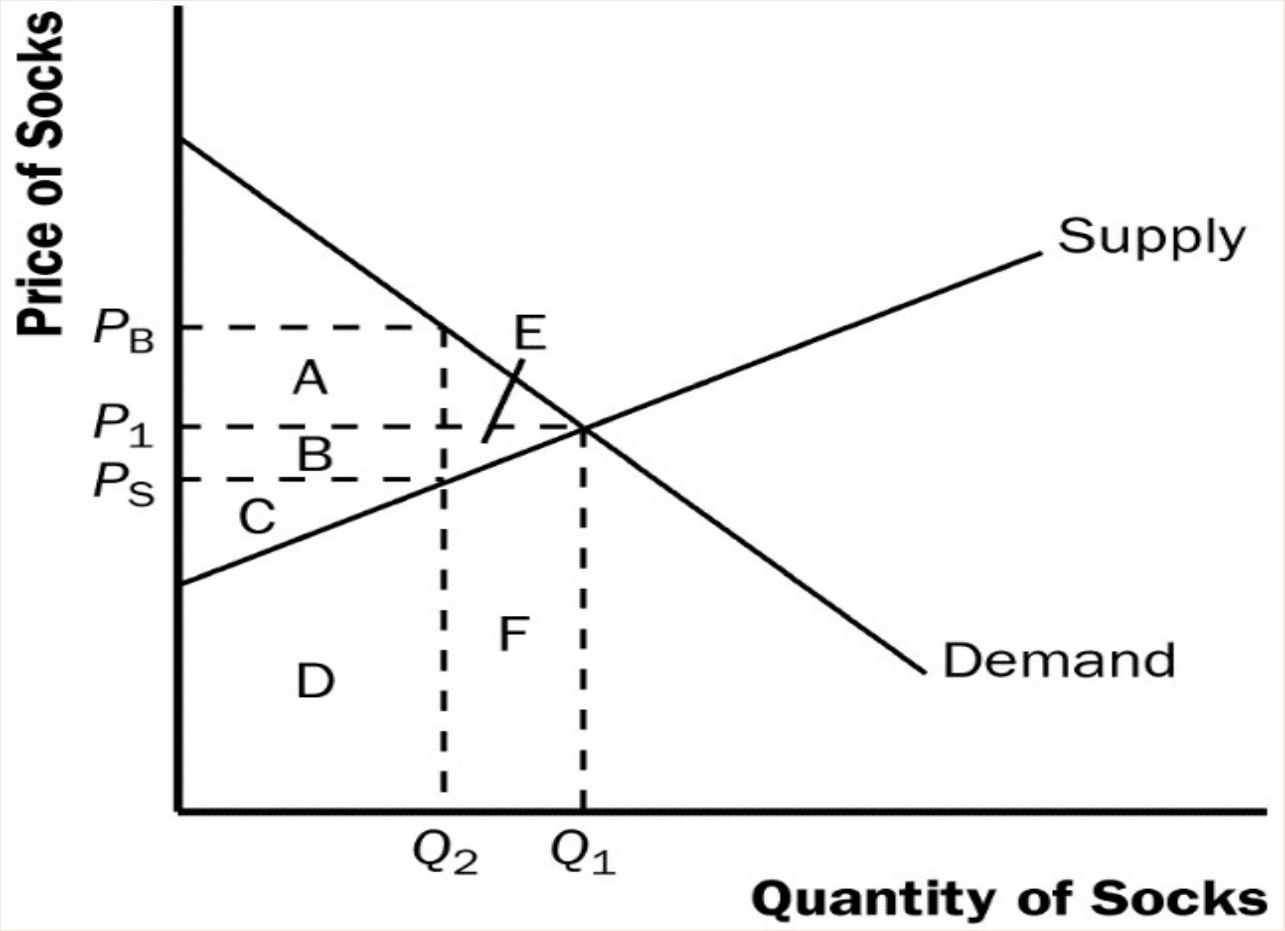
Appendix Question

- After economics class one day, your friend suggest that taxing food would be a good way to raise revenue because the demand for food is quite inelastic.
  - In what sense is taxing food a “good” way to raise revenue?
  - In what sense is it not a “good” way to raise revenue?

Appendix Question ✶✶✶

- The government places a tax on the purchase of socks 假設稅金為$T
  - Illustrate the effect of this tax on equilibrium price and quantity in the sock market.
    - Identify the following areas both before and after imposition of the tax
      - **total spending (α)** by consumers
      - **total revenue (β)** for producers
      - **government tax revenue (γ)**.
  - Does **the price received by producers** rise or fall? Can you tell whether **total receipts for producers** rise or fall?
  - Does **the price paid by consumers** rise or fall? Can you tell whether **total spending by consumers** rises or falls? If total consumer spending falls, does **consumer surplus** rise?

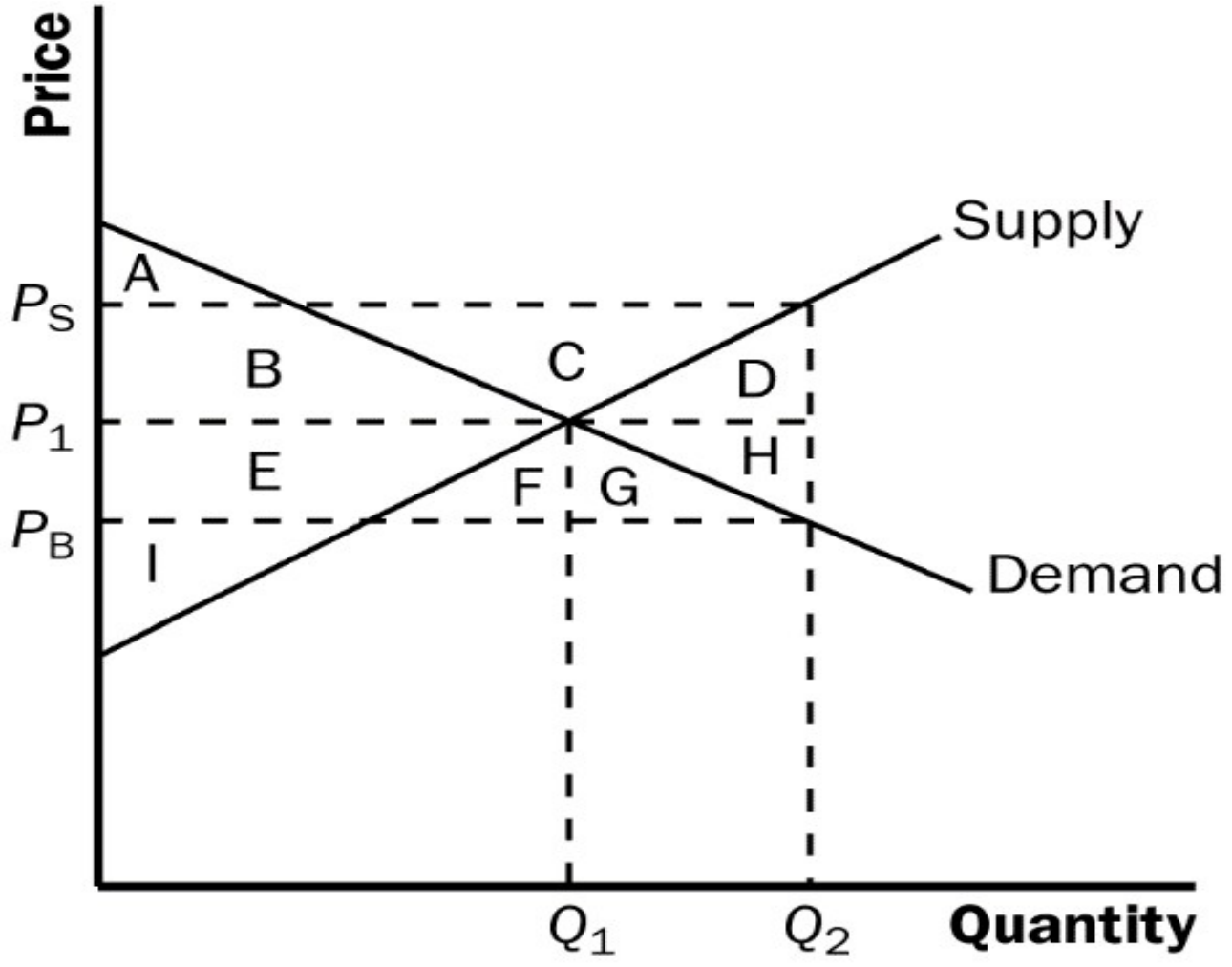
Reference


Price of Socks
P_B
P_1
P_S
A
B
C
D
E
F
Supply
Demand
Q_2
Q_1
Quantity of Socks

Appendix Question

- This chapter analyzed the welfare effects of a tax on a good. Now consider the opposite policy.
- Suppose that the government subsidizes a good: For each unit of the good sold, the government *pays $2 to the buyer*.
- How does this subsidy affect consumer surplus , producer surplus, tax revenue, and total surplus? Does a subsidy lead to **a deadweight loss**?

Appendix Reference





How Big Should the Government Be?



- A bigger government
  - Provides more services, but requires higher taxes, which cause DWLs
  - The larger the DWL from taxation, the greater the argument for smaller government
- Tax on labor income - especially important
  - Biggest source of government revenue
- Marginal tax rate: about 40%
- How big is the DWL?

CASE STUDY

How Big Should the Government Be?

- 40% labor tax - Small or large deadweight loss?



"What's your position on the elasticity of labor supply?"

- Some believe labor supply is fairly inelastic
  - Almost vertical: most people would work full-time regardless of wage
  - Tax on labor: small DWL



How Big Should the Government Be?

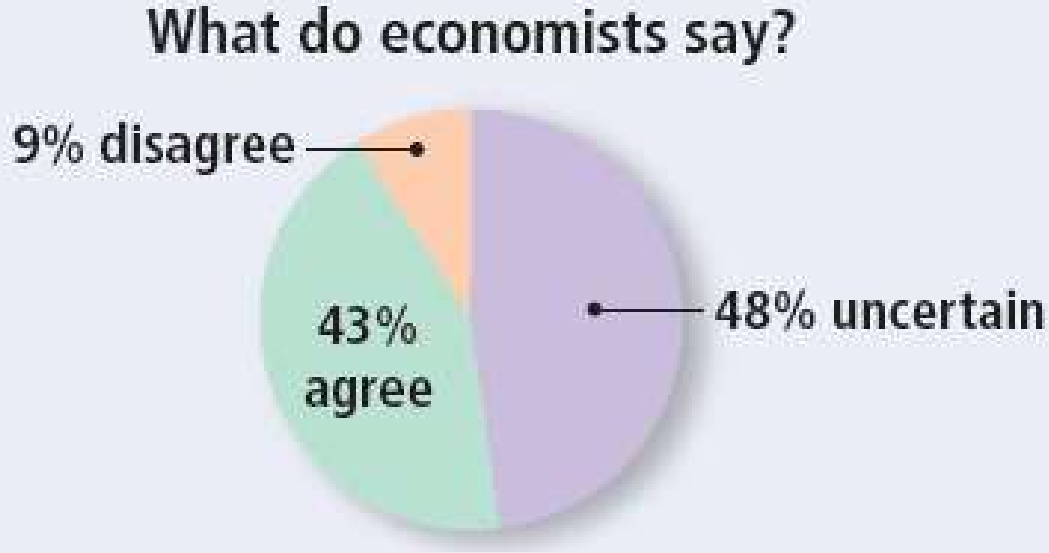


- Others: labor supply is more elastic
  - Labor taxes are highly distorting: some groups of workers have elastic supply and can respond to incentives
  - Tax on labor: greater DWL
    - Many workers can adjust their hours
    - Some families have 2nd earners; some discretion over whether and how much to work
    - Many of the elderly can choose when to retire
    - Some people work in the "underground economy" to evade high taxes

ASK THE EXPERTS

The Laffer Curve 拉佛曲線;拉法爾曲線

"A cut in federal income tax rates in the United States right now would lead to higher national income within five years than without the tax cut."



National income (國民所得) means **the value of goods and services produced by a country during a financial year**



Effects of Changing the Size of the Tax

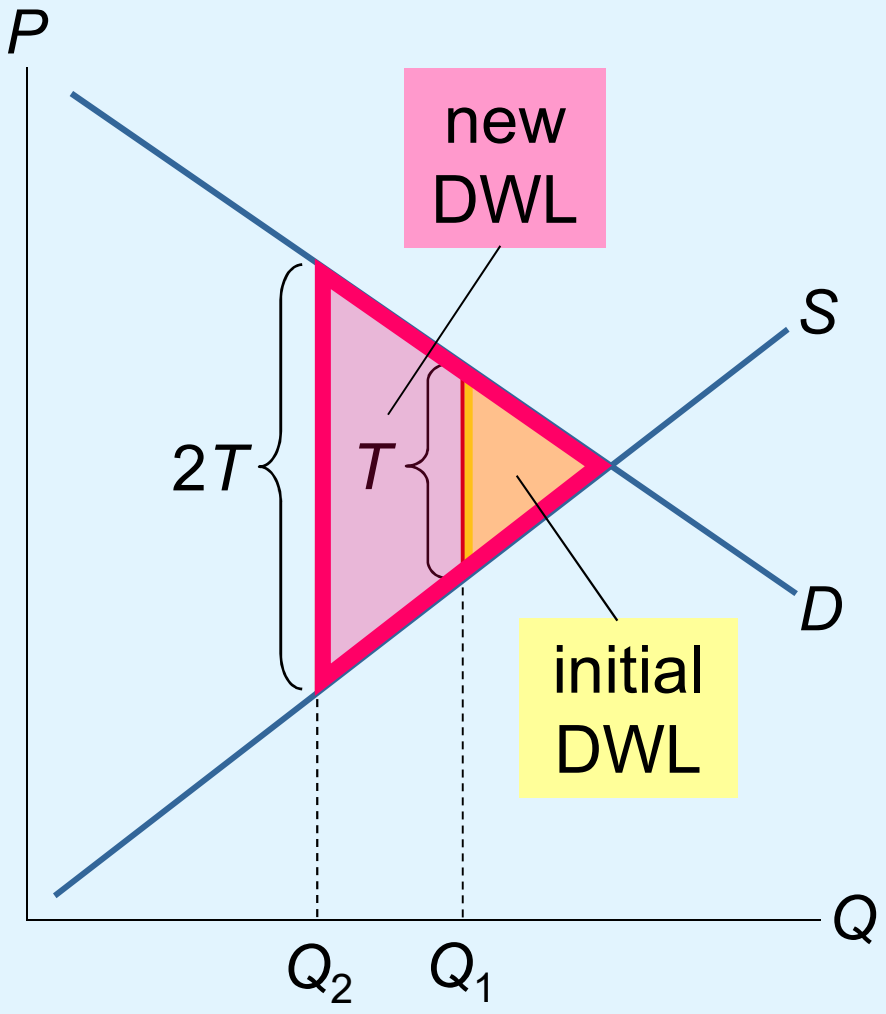
- As the tax increases
  - Deadweight loss increases
    - Even more rapidly than the size of the tax
  - Tax revenue
    - Increases initially
    - Then decreases
    - The higher tax: drastically reduces the size of the market

DWL and the Size of the Tax

Initially, the tax is T per unit.

Doubling the tax

causes the DWL to more than double.

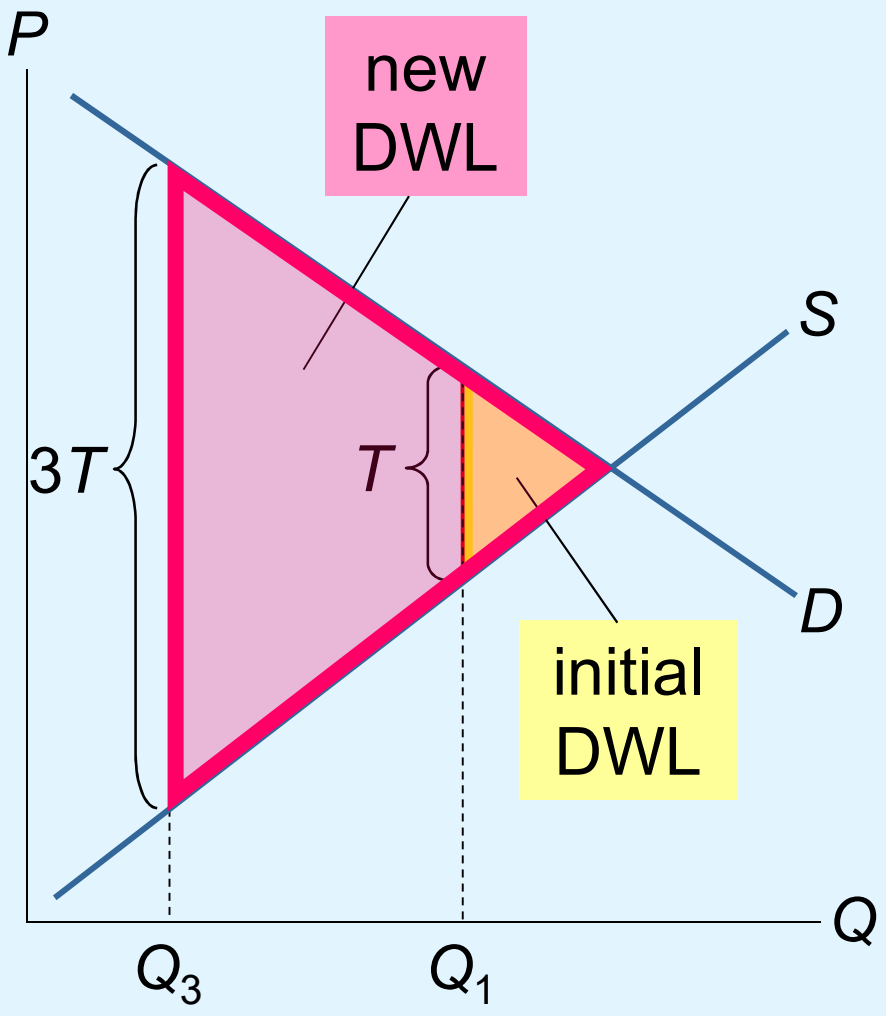


DWL and the Size of the Tax

Initially, the tax is T per unit.

Tripling the tax

causes the DWL to more than triple.



DWL and the Size of the Tax

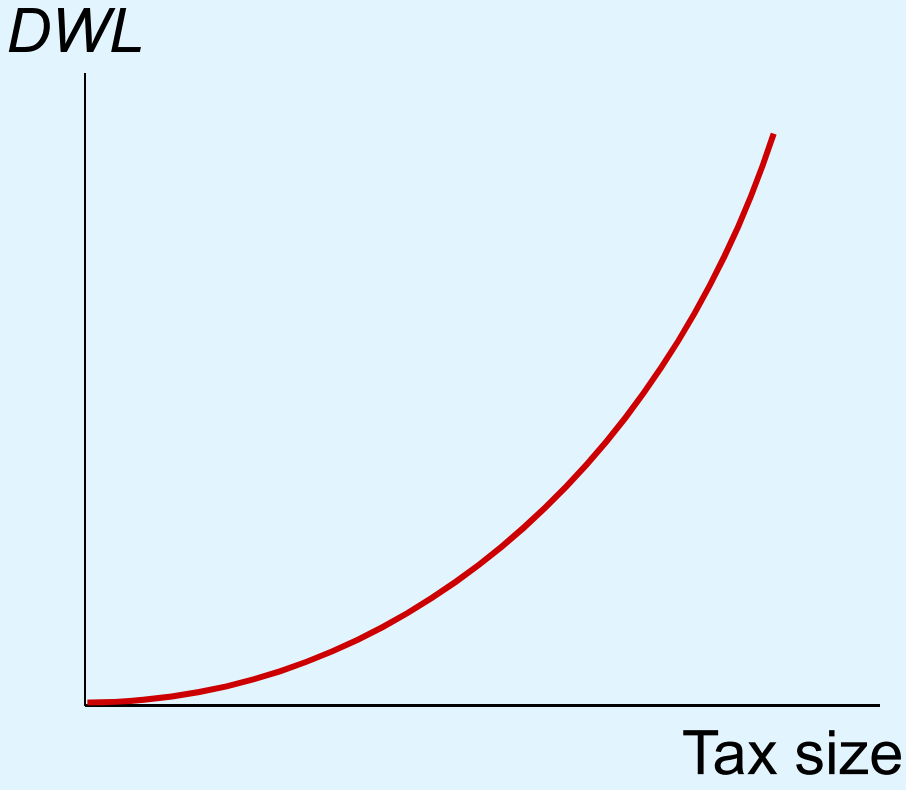
Implication

When tax rates are low, raising them doesn't cause much harm, and lowering them doesn't bring much benefit.

When tax rates are high, raising them is very harmful, and cutting them is very beneficial.

Summary

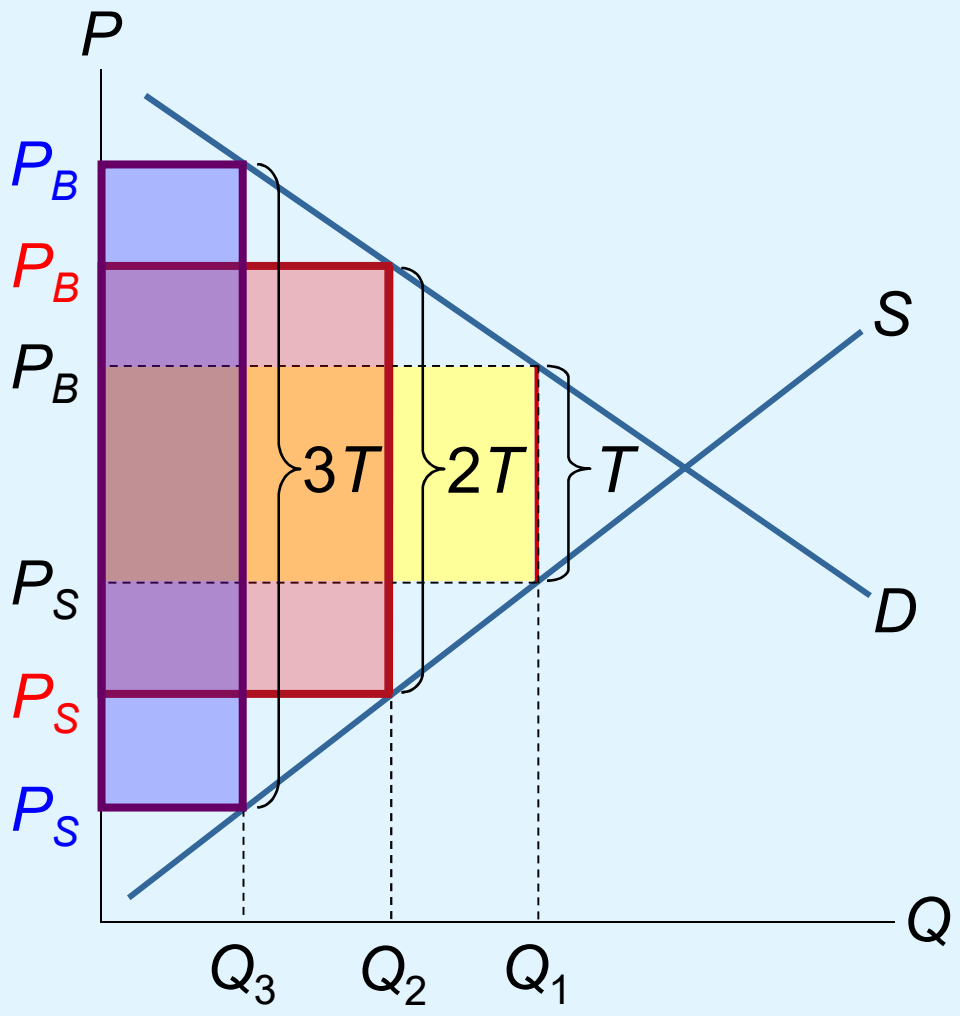
When a tax increases, DWL rises even more.



Revenue and the Size of the Tax

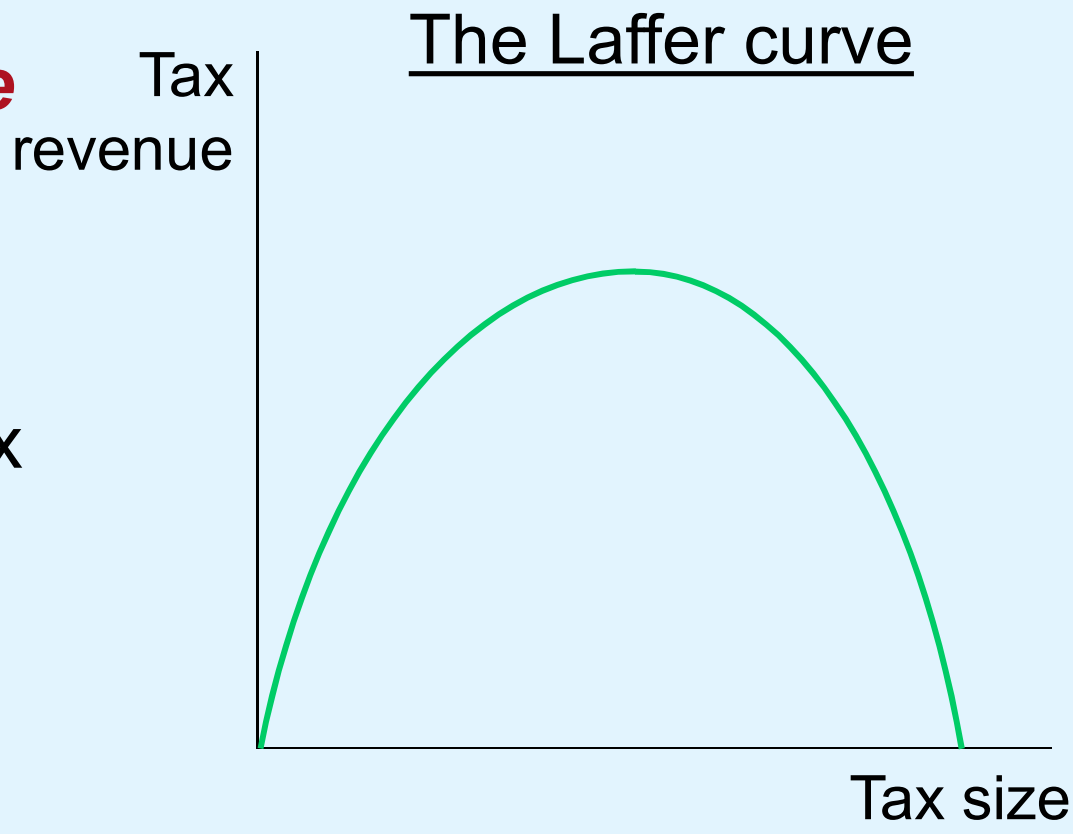
When the tax is small, increasing it causes tax revenue to rise.

When the tax is larger, increasing it causes tax revenue to fall.



Revenue and the Size of the Tax

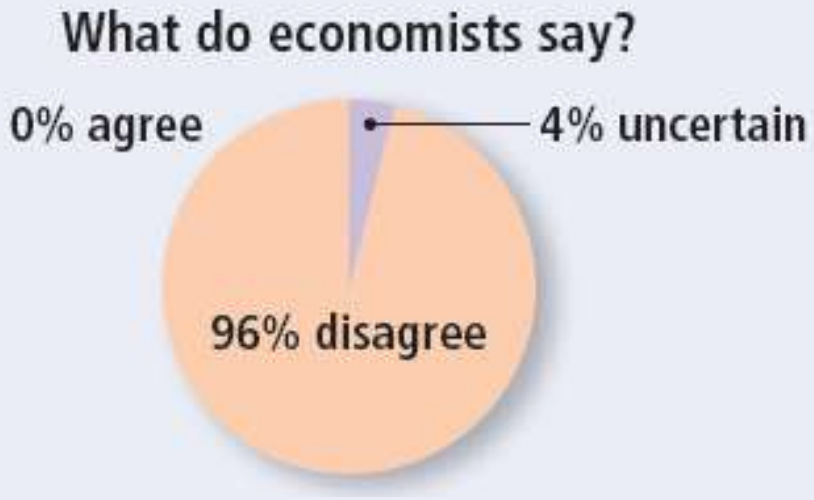
The ***Laffer curve*** shows the relationship between the size of the tax and tax revenue.



ASK THE EXPERTS

The Laffer Curve

"A cut in federal income tax rates in the United States right now would raise taxable income enough so that the annual total tax revenue would be higher within five years than without the tax cut."



Q: Can Tax Cuts Increase Tax Revenue?

Summary

- A tax on a good reduces the welfare of buyers and sellers. This welfare loss usually exceeds the revenue the tax raises for the govt.
- The fall in total surplus (consumer surplus, producer surplus, and tax revenue) is called the deadweight loss (DWL) of the tax.
- A tax has a DWL because it causes consumers to buy less and producers to sell less, thus shrinking the market below the level that maximizes total surplus.

Summary

- The price elasticities of demand and supply measure how much buyers and sellers respond to price changes. Therefore, higher elasticities imply higher DWLs.
- An increase in the size of a tax causes the DWL to rise even more.
- An increase in the size of a tax causes revenue to rise at first, but eventually revenue falls because the tax reduces the size of the market.

Appendix Question

- Hotel rooms in Smalltown go for $100, and 1,000 rooms are rented on a typical day
  - To raise revenue, the mayor decides to charge hotels a tax of $10 per rented room.
    - After the tax is imposed, the going rate for hotel rooms rises to $108, and the number of rooms rented falls to 900.
    - Calculate the amount of revenue this tax raises for Small town and the deadweight loss of the tax.
  - The mayor now doubles the tax to $20.
    - The price rises to $116, and the number of rooms rented falls to 800.
    - Calculate tax revenue and deadweight loss with this larger tax.
    - Are they double, more than double, or less than double?

Question

✶✶✶

- Suppose that a market is described by the following supply and demand equations: $Q^S = 2P$, $Q^D = 300 - P$
- Solve the equilibrium price and the equilibrium quantity
- Suppose that a tax of T is placed on buyers, so the new demand equation is $Q^D = 300 - (P+T)$. Solve for the new equilibrium.
- Tax revenue is T x Q. Use your answer to solve for tax revenue as a function of T. *Graph this relationship for T between 0 and 300*.
- Solve for deadweight loss as a function of T. *Graph this relationship for T between 0 and 300*.
- The government now levies a tax of $200 per unit of this good. Is this a good policy? Why or why not? Can you propose a better policy?

Does Congress Decide Who Pays the Taxes?



Eye on Congress (1 of 4)

Does congress Decide Who pays the Taxes?

Congress says that employers and workers pay the same Social Security tax contributions (7.65 percent each in 2015).

But because the elasticity of demand for labor is much greater than the elasticity of supply of labor, workers end up paying most of the Social Security tax.

But there is one thing that Congress can do to influence who pays a tax.

It can pass a tax law (or tax rebate law) that doesn't impact the margin on which decisions turn.

Recently, Congress passed such a law.

Eye on Congress (2 of 4)

Does congress Decide Who pays the Taxes?

On February 17, 2009, the President signed the American Recovery and Reinvestment Act (美國復甦與再投資法).

Among the Act's many provisions is a "Making Work Pay" tax credit of $400 for a single worker and $800 for a couple.

A tax credit is a fixed reduction in the amount paid in personal income tax (in the current case, $400).

For most people, a tax credit has no effect on their supply of labor.

The tax credit doesn't influence the work-hours choice.

Eye on Congress (3 of 4)

Does congress Decide Who pays the Taxes?

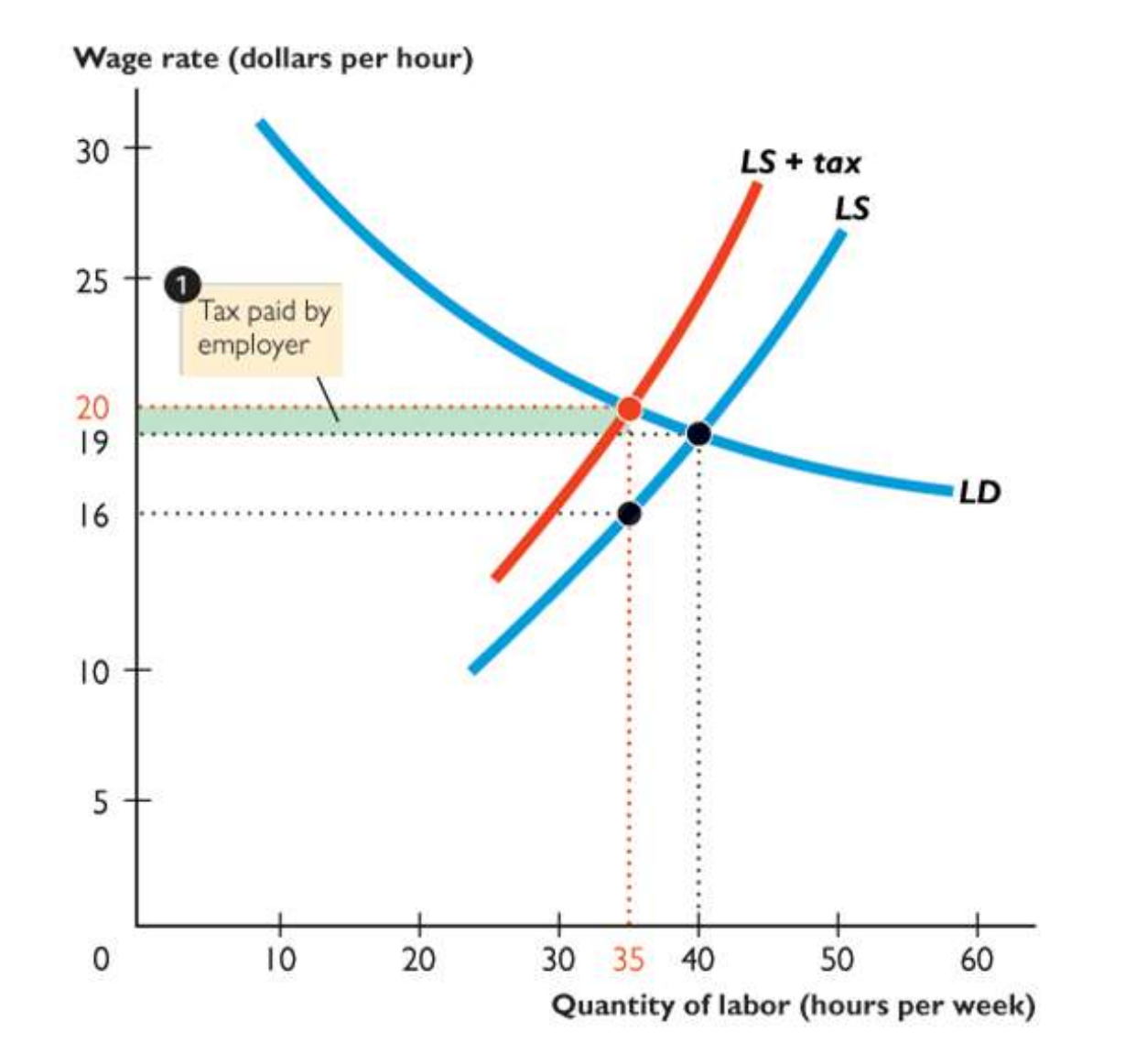
A 20% income tax rate shifts the labor supply curve from *LS* to *LS* + *tax*.

The pre-tax wage rises by $1 to $20 per hour,

the after-tax wage falls by $3 to $16 per hour, and

the workweek falls from 40 hours to 35 hours.

Workers pay 75% of the tax and employers pay 25%.



Eye on Congress (4 of 4)

Does congress Decide Who pays the Taxes?

Suppose that Congress now passes an Act that gives workers a tax credit of $30 a week.

Tax paid by workers falls.

Marginal tax rate remains at 20%, but now workers pay only 68% of the tax and employers pay 32%.

Congress has worked around the elasticities!

資方35/(35+75)=32%
勞方75/(35+75)=68%

